Chapter 4

Single-Cell Cytokine Analysis to Characterize CAR-T Cell Activation

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Abstract

With the increase in the implementation of adoptive transfer anti-CD19 chimeric antigen receptor (CAR) T cell therapy, we introduce a novel platform to study their heterogeneous cytokine response at the single-cell level following activation. Here, we describe an activation platform which incubates single CAR-T cells with single-target CD19 antigen-presenting cells (APCs) and measures the resulting cytokine secretions. This can be compared to the more traditional bulk activation modalities. Both pipelines of activation are measured using our high-throughput, multiplexed single-cell secretomic microchip composed of an antibody barcode array and subnanoliter microchambers, capable of detecting up to 42 unique proteins. This platform has also been adapted to cell lines as well as primary cells.

Key words Single-cell omics, Single-cell cytokine profiling, Immunophenotyping, Functional heterogeneity, High throughput, Multiplexed, Cytokine secretion, Activation, Anti-CD19 CAR-T cells

1 Introduction

In recent years, there have been a rise in the efficacy of adoptive transfer anti-CD19 chimeric antigen receptor (CAR) T cell therapy in hematological malignancies such as chronic lymphocytic leukemia (CLL), B cell acute lymphoblastic leukemia (B-ALL), and pediatric acute lymphoblastic leukemia (ALL). With this, there remains a need for more effective platforms to fully understand their functional heterogeneity following activation [1–3]. Common platforms used for cytokine analysis include fluorescent-activated cell sorting (FACS), enzyme-linked immunosorbent assays (ELISA), and ELISpot/FluoroSpot, each with inherent limitations. FACS uses surface markers and intracellular proteins to investigate cellular characteristics. However, this does not directly measure the actual secreted proteins and can lead to the overestimation of cytokine-secreting cells [4]. Further, only a limited number of intracellular cytokines and chemokines can be measured, thereby preventing a

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comprehensive view of CAR-T polyfunctionality [5]. ELISA, a microtiter plate immunoassay, has a limited multiplexing capability and sensitivity. Additionally, this platform requires a large sample volume and analysis time [6, 7]. Alternatively, the ELISpot/FluoroSpot method allows for the direct analysis of cytokine secretions but again falters in its multiplexing capabilities [8]. Given the need for a higher-throughput and highly multiplexed immunophenotyping technology, we have developed a single-cell cytokine secretion microchip that integrates an antibody barcode array and subnanoliter microchambers to detect up to 42 unique proteins [9, 10]. This modality allows for the deep functional profiling of anti-CD19 CAR-T cells, imperative in fully understanding their heterogeneity and polyfunctionality [5].

Not only do multiple platforms exist for the measurement of cytokine secretion, but also multiple methodologies exist for the activation of anti-CD19 CAR-T cells. One method uses soluble ligands, such as transforming growth factor beta (TGF-β), CD19 ectodomain, and GFP variants [11]. However, these only allow for an assessment of cytokine secretion at the populational level. Another platform utilizes CD19-coated beads to stimulate CD19–BB-3z CAR-T cells for subsequent measurement of cytokine output, but these beads are not commercially available [5]. Finally, the use of CD19 engineered antigen-presenting cells (APCs) for anti-CD19 CAR-T cell activation has been widely implemented. This strategy has facilitated the correlative study between stimulated pre-infusion CAR-T cell polyfunctionality, CAR-T cell expansion in vivo, and patient response [12]. Our platform uses these CD19 APCs to stimulate anti-CD19 CAR-T cells in two ways (Fig. 1). The first methodology investigates their cell–cell interactions following co-incubation by magnetically separating the CAR-T cells and APCs and performing single-cell cytokine analysis of the CAR-T cells. The latter, novel activation platform incubates a single CAR-T cell with a single-target APC within the same microchamber and measures the resulting cytokines. Here, we outline these activation platforms in tandem with our protein secretion methodology. For the purpose of this protocol, we focus on CAR-T cells. However, this platform has the potential to be adapted to other immunomodulatory therapies such as tumor-infiltrating lymphocytes.

2 Materials

2.1 CAR-T Cell Preparation and Activation

2.1.1 CAR-T Cell Culture

1. Anti-CD19 scFv–CD28–4-1BB–CD3-ζ T cells (Promab).
2. Complete X-VIVO 10 media.
3. IL-2 (10 ng/mL).
4. Anti-CD4 FITC.
5. Anti-CD8 Alexa Fluor 647.

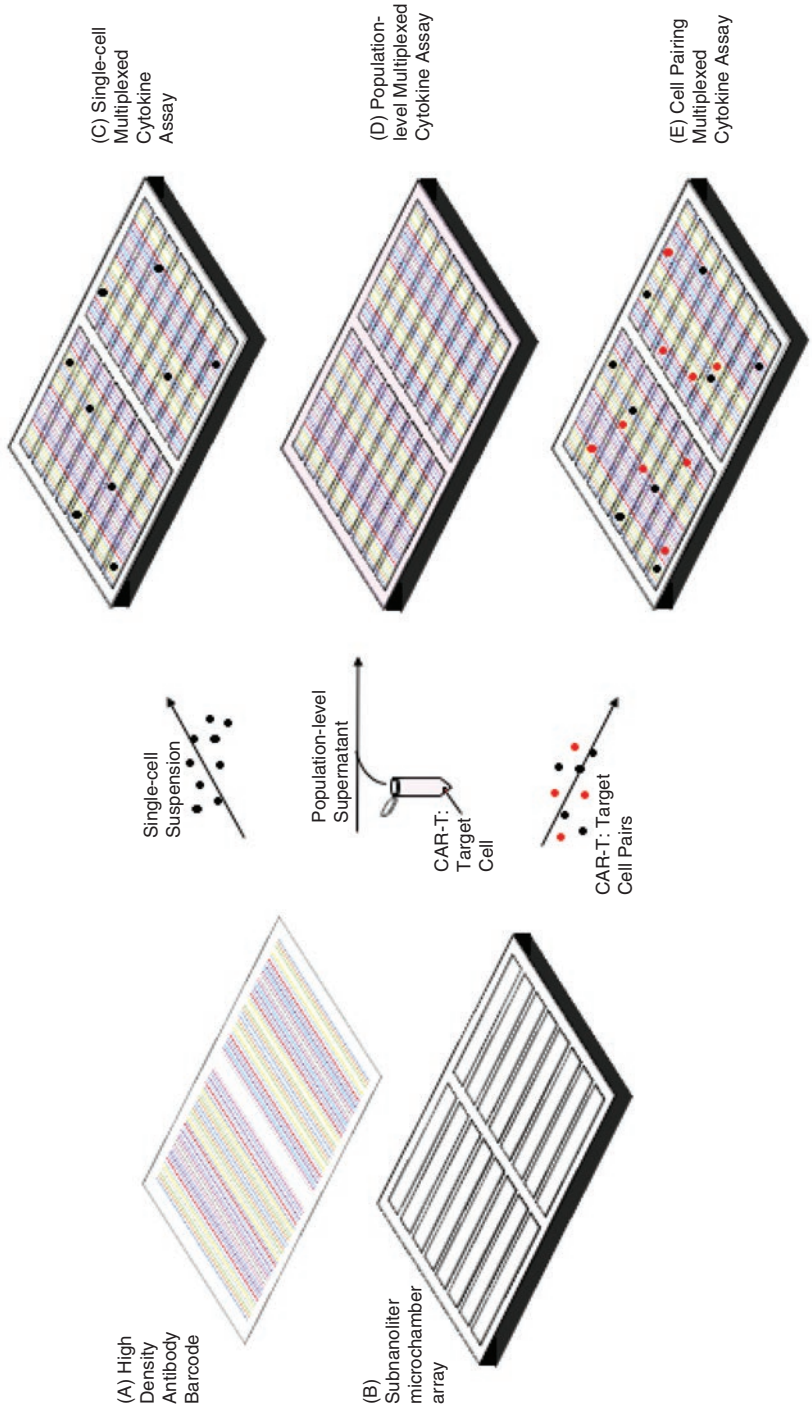

Fig. 1 Multiplexed cytokine analysis for CAR-T cell therapy. Illustration of the (**a**) high-density antibody barcode chip and the (**b**) subnanoliter microchamber array chip for a high-throughput multiplexed cytokine secretion assay at the (**c**) single-cell, (**d**) population, and (**e**) cell-pair levels

6. Phosphate-buffered solution (PBS).
7. RPMI medium supplemented with 10% FBS (*see* **Note 1**).

*2.1.2 Target Cell Culture (**See** Note 2)*

1. Vybrant DiD.
2. RPMI medium.
3. RPMI medium supplemented with 10% fetal bovine serum (FBS).

2.2 Device Preparation for Cytokine Analysis

2.2.1 Silicon Master Mold

1. Patterned photomask (*see* **Note 3**).
2. 4 in. p-Type CZ silicon wafer (Silicon Quest Int'l).
3. SPR 220 i-line 1.5 (Dow).
4. MF-319 Microposit developer (Shipley).
5. Chlorotrimethylsilane (TMCS) (Sigma-Aldrich).
6. SU-8 negative photoresist (MicroChem).
7. SU-8 developer (MicroChem).
8. 100% Isopropanol.

2.2.2 Barcode Flow Patterning and Cell Capture Microchamber Chip Fabrication

1. Silicon master mold (barcode flow patterning and cell capture microchamber).
2. Aluminum foil.
3. Polydimethylsiloxane (PDMS) RTV615 A (siloxane/potting agent), RTV615 B (cross-linking agent) (Momentive) (*see* **Note 4**).
4. Desiccator.
5. 70% Ethanol.
6. 100% Isopropanol.
7. DI water.
8. PDMS hole puncher.
9. Magic Tape 810 (*see* **Note 5**).

2.3 Barcode Chip Fabrication

1. SuperChip microarray poly-L-lysine glass slide (Thermo Scientific).
2. PDMS flow-patterning chip.
3. 23G Metal pins, blunt ends.
4. Microbore tubing 0.02 in. (Tygon).
5. 23G Luer-Stub Adapter (Becton Dickinson).
6. Nitrogen gas.
7. Purified capture monoclonal antibodies, ELISA compatible.
8. Phosphate-buffered solution (PBS).
9. Blocking solution: 3% bovine serum albumin (BSA) in PBS (*see* **Note 6**).
10. DI water.

2.4 Cytokine Assay

1. PDMS cell capture microchamber array chip.
2. PDMS flow-patterning chip (barcode chip).
3. Culture media (*see* **Note 7**).
4. Cell suspension.
5. Plexiglass clamps.
6. 18-8 Stainless steel socket head cap screw 4-40 thread, 5/8″ length.
7. Ballpoint L-key 3/32″ Hex.
8. Blocking solution: 3% BSA in PBS.
9. Biotinylated antibody, ELISA compatible (*see* **Note 8**).
10. Streptavidin–fluorophores.
11. Microscope with automatic image stitching (*see* **Note 9**).
12. Genepix array scanner (Molecular Devices).

3 Methods

All procedures were conducted at room temperature unless otherwise specified.

3.1 CAR-T Cell Preparation and Activation

3.1.1 CAR-T Cell Culture

1. Culture human CAR-T cells at 37 °C; 5% CO_2 in complete X-VIVO 10 media supplemented with IL-2 at a concentration of 0.5×10^6 cells/mL.
2. Label CD4/CD8 subsets using anti-CD4 FITC and anti-CD8 Alexa Fluor 647 for 15 min, rinse 2× with PBS and 1× in target cell medium (*see* **Notes 2** and **10**).

3.1.2 Target Cell Culture

1. Culture target cells following manufacturing instructions in respective media (*see* **Note 2**).
2. To label, resuspend cells at a final concentration of 1×10^6 cells/mL in serum-free media with Vybrant DiD (5 μL/mL of cells) for 20 min at 37 °C; 5% CO_2 (*see* **Note 11**).
3. Wash labeled cells three times in respective medium (*see* **Note 2**).

3.2 Silicon Master Mold Fabrication

This procedure should be performed in the cleanroom.

3.2.1 Fabrication of Flow-Patterning Master Mold

1. Heat hot plates to 120 °C.
2. Bake a new silicon wafer on the preheated hot plate for 5 min.
3. Place wafer at the center of the photoresist spinner's wafer holder. Fix its position with the vacuum pressure of the wafer holder.

4. Use nitrogen air to remove any dust or particles.
5. For the photoresist coat, apply photoresist SPR 220 i-line 1.5 (*see* **Note 12**) on the wafer (4 in.), and spin-coat at 900 rpm for 5 s and then at 3000 g for 1 min (*see* **Notes 13** and **14**).
6. Place coated wafer on 100 °C hot plate for 3 min to soft bake.
7. Decrease the hot plate temperature to 55 °C at a rate of 5 °C/min (*see* **Note 15**).
8. Expose wafer to 150 mJ/cm^2 UV light using EVG 620 contact/proximity mask aligner (*see* **Note 16**).
9. Bake wafer on a 120 °C hotplate for 3 min or more, and gradually cool down to room temperature at a rate of 5 °C/min (*see* **Note 15**).
10. Develop wafer with Microposit MF-319 developer for approximately 1 min.
11. Use dry reactive-ion etching (DRIE) for wafer pattern etching.
12. Following DRIE, treat with TMCS via vapor deposition in a vacuum chamber for 20 min.

3.2.2 Fabrication of Single-Cell Capture Microchamber Master Mold

1. Heat hot plates to 120 °C.
2. Bake a new silicon wafer on the preheated hot plate for 5 min.
3. Place wafer at the center of the photoresist spinner's wafer holder. Fix its position with the vacuum pressure of the wafer holder.
4. Use nitrogen air to remove any dust or particles.
5. For the photoresist coat, apply SU-8 negative photoresist on the wafer, and spin-coat at 500 rpm for 20 s and then 2000 g for 60 s (*see* **Notes 13** and **14**).
6. To soft bake, place coated wafer on a 65 °C hot plate, and increase the temperature to 95 °C at a rate of 5 °C/min. Then bake for 15 min at 95 °C.
7. Decrease the hot plate temperature to 55 °C at a rate of 5 °C/min (*see* **Note 15**).
8. Expose the wafer employing the EVG 620 contact/proximity mast aligner using the following parameters: soft contact/exposure time = 3 s/interval = 20 s/separation = 10 cycles.
9. Repeat **steps 7** and **8**.
10. Develop wafer in a glass dish filled of SU-8 developer for 7–8 min with constant, gentle shaking.
11. Clean the wafer with IPA, and air-dry to remove excess liquid on the wafer surface.

3.3 Barcode Flow-Patterning Chip Device Preparation (See Note 17)

1. Vigorously mix PDMS siloxane/potting agent with the PDMS cross-linking agent in a 10:1 ratio (*see* **Note 18**).
2. Remove any dust or particles from the flow-patterning silicon master mold using filtered air (*see* **Note 19**).
3. Wrap the wafer in an aluminum foil boat (*see* **Note 20**).
4. Pour PDMS mix (50 g) over the flow-patterning silicon master mold. Degas in a desiccator for 1 h to remove all air bubbles from the PDMS.
5. Transfer the mold to a 60 °C oven, and bake for 4 h (*see* **Note 21**).
6. Remove the mold, and allow to cool to room temperature.
7. Separate the cured PDMS from silicon wafer, and keep it clean and dust free (*see* **Note 22**).
8. Cut the patterned area out using a razor blade, ensuring the final product is proportionate to the poly-L-lysine glass slide.
9. Using the PDMS hole puncher, cut inlet and outlet holes for the antibodies (*see* **Note 23**).
10. Sonicate the PDMS chip in 70% ethanol for 20 min. Repeat.
11. Sonicate the PDMS chip in 100% isopropanol for 20 min.
12. Dry the surface of the chip with filtered air, and place in an 80 °C oven for 2 h to remove all liquids (*see* **Note 24**).
13. Cover the patterned area with Scotch tape. After applying firm pressure, remove tape, and repeat until the PDMS is free of dust and particles (*see* **Notes 3** and **25**).
14. Remove any dust or particles from the poly-L-lysine-coated glass slide using filtered air, and attach to the PDMS barcode flow-patterning chip (*see* **Note 26**).
15. Using a standard microscope, analyze channels section by section. Check for debris, dust, cross talk, air bubbles, deformities, or blockages (*see* **Note 27**).
16. Thermal bond the glass slide to the PDMS chip in a 75 °C oven for 2 h, and allow to cool to room temperature (*see* **Note 28**). This bonded PDMS and glass slide will be referenced as the *Flowpattern Assembly* in the proceeding steps.

3.4 Cell Capture Microchamber Chip Device Preparation

1. Follow **steps 1–7** from Subheading 3.2, now using the Cell Capture Microchamber Chip Silicon Master Mold.
2. Cut the patterned area out using a razor blade, ensuring the final product is proportionate to the barcode area of the flow-patterning chip.
3. Store microchamber chip in a dust-free container until ready to utilize.

3.5 Flow Patterning of Barcode Chip

1. With Luer-Stub Adapters attached to tubing, insert hollow pins at the alternate end.
2. Load 2 μL of primary antibody into the pin, and insert the pin into the inlet hole of the *Flowpattern Assembly* (*see* **Note 29**).
3. Attach the Luer-Stub Adapter to pressure-regulated nitrogen gas.
4. Repeat for all antibodies required for the cytokine analysis.
5. Using the pressure regulator, slowly open nitrogen flow to 2 PSI (*see* **Note 30**).
6. Incubate chip overnight in a clean enclosed dark space (*see* **Note 31**).
7. Turn off the nitrogen flow, and remove pins from the chip (*see* **Note 32**).
8. Submerge *Flowpattern Assembly* in 5% BSA, and carefully remove the PDMS chip from the glass barcode chip. Set PDMS chip aside to clean for later reuse (*see* **Note 33**).
9. Remove the glass barcode chip from the solution, and wash 3× with 1 mL of 5% BSA.
10. Block the surface of the barcode chip with ~1 mL of 5% BSA in a clean enclosed dark space for 1 h (*see* **Note 31**).
11. Wash with PBS, followed by (1:1) PBS/DI water, and then DI water (*see* **Note 32**).
12. Using filtered compressed air, blow the surface of the barcode chip dry (*see* **Note 34**).
13. Store at 4 °C until use (*see* **Note 35**).

3.6 Cytokine Assay

3.6.1 Population-Level Multiplexed Cytokine Assay of CAR-T: Target Cell Pairings

1. Co-incubate CAR-T cells with target cells in a 1:1 ratio and CAR-T cells and target cells alone, each at a 100,000 cells/mL concentration (*see* **Note 36**).
2. Incubate for 10 h at 37 °C; 5% CO_2.
3. Following incubation, pellet cells at 200 × *g*, and save the supernatant.
4. Sonicate Cell Capture Microchamber Chip in 5% BSA for 20 min.
5. Oxygen-plasma treat the Cell Capture Microchamber Chip, and top with culture media (*see* **Note 37**).
6. Place Cell Capture Microchamber Chip on the bottom clamp plate.
7. Pipette 40 μL of supernatant onto cell capture area evenly, and let cells settle for at least 2 min (*see* **Note 38**).
8. Add ~40 μL of media along the top of the Cell Capture Microchamber Chip (*see* **Note 37**).

9. With the antibody barcode facing the microchambers, place the glass barcode chip lengthwise onto the Cell Capture Microchamber Chip (*see* **Note 39**).
10. Top with the clamp plate and screw device together evenly (*see* **Note 40**).
11. Image chip with a microscope with automatic image stitching (*see* **Note 41**), and incubate at 37 °C and 5% CO_2 for 12–24 h.
12. Submerge assembly into 5% BSA, disassemble the clamp plates, and carefully remove the barcode chip from the Cell Capture Microchamber Chip (*see* **Note 42**).
13. Rinse chip 5× with 1 mL of 5% BSA (*see* **Note 43**).
14. In a dark flat environment, add 400 μL of secondary antibody (1:200 dilution in 5% BSA), and incubate for 40 min (*see* **Note 31**).
15. Following incubation, repeat **step 10**.
16. In a dark flat environment, add 400 μL of streptavidin–fluorophore (1:100 dilution in 5% BSA), and incubate for 30 min (*see* **Note 31**).
17. Following incubation, repeat **step 10**.
18. In a dark flat environment, block chip with 5% BSA, and incubate for 30 min (*see* **Note 31**).
19. Wash with PBS, followed by (1:1) PBS/DI water, and then DI water (*see* **Note 32**).
20. Using filtered compressed air, blow the surface of the barcode chip dry (*see* **Note 34**).
21. Scan chip using Genepix array scanner (*see* **Note 44**).
22. Combine Genepix array scanner with the positions of the microchambers and cell counts.

3.6.2 Single-Cell Multiplexed Cytokine Assay

1. Sonicate Cell Capture Microchamber Chip in 5% BSA for 20 min.
2. Oxygen-plasma treat the Cell Capture Microchamber Chip, and top with culture media (*see* **Note 37**).
3. Place Cell Capture Microchamber Chip on the bottom clamp plate.
4. Pipette 200 μL of 0.3 million cells/mL cell suspension onto cell capture area evenly, and let cells settle for at least 2 min (*see* **Note 38**).
5. Add ~40 μL of media along the top of the Cell Capture Microchamber Chip (*see* **Note 37**).
6. With the antibody barcode facing the microchambers, place the glass barcode chip lengthwise onto the Cell Capture Microchamber Chip (*see* **Note 39**).

7. Top with the clamp plate and screw device together evenly (*see* **Note 40**).
8. Image chip with a microscope with automatic image stitching (*see* **Note 41**), and incubate at 37 °C for 12–24 h.
9. Submerge assembly into 5% BSA, disassemble the clamp plates, and carefully remove the barcode chip from the Cell Capture Microchamber Chip (*see* **Note 42**).
10. Rinse chip 5× with 1 mL of 5% BSA (*see* **Note 43**).
11. In a dark flat environment, add 400 μL of secondary antibody (1:200 dilution in 5% BSA), and incubate for 40 min (*see* **Note 31**).
12. Following incubation, repeat **step 10**.
13. In a dark flat environment, add 400 μL of streptavidin–fluorophore (1:100 dilution in 5% BSA), and incubate for 30 min (*see* **Note 31**).
14. Following incubation, repeat **step 10**.
15. In a dark flat environment, block chip with 5% BSA, and incubate for 30 min (*see* **Note 31**).
16. Wash with PBS, followed by (1:1) PBS/DI water, and then DI water (*see* **Note 32**).
17. Using filtered compressed air, blow the surface of the barcode chip dry (*see* **Note 34**).
18. Scan chip using Genepix array scanner (*see* **Note 44**).
19. Combine Genepix array scanner with the positions of the microchambers and cell counts (Fig. 2).

3.6.3 Cell-Pairing Multiplexed Cytokine Assay

1. After labeling CAR and cells with membrane dye, incubate cells in a 1:1 ratio for 15 min in culture medium (*see* **Note 7**).
2. Sonicate Cell Capture Microchamber Chip in 5% BSA for 20 min.
3. Oxygen-plasma treat the Cell Capture Microchamber Chip, and top with culture media (*see* **Note 37**).
4. Place Cell Capture Microchamber Chip (Fig. 2) on the bottom clamp plate.
5. Pipette 200 μL of 0.3 million cells/mL cell suspension onto cell capture area evenly, and let cells settle for at least 2 min (*see* **Note 38**).
6. Add ~40 μL of media along the top of the Cell Capture Microchamber Chip (*see* **Note 37**).
7. With the antibody barcode facing the microchambers, place the glass barcode chip lengthwise onto the Cell Capture Microchamber Chip (*see* **Note 39**).

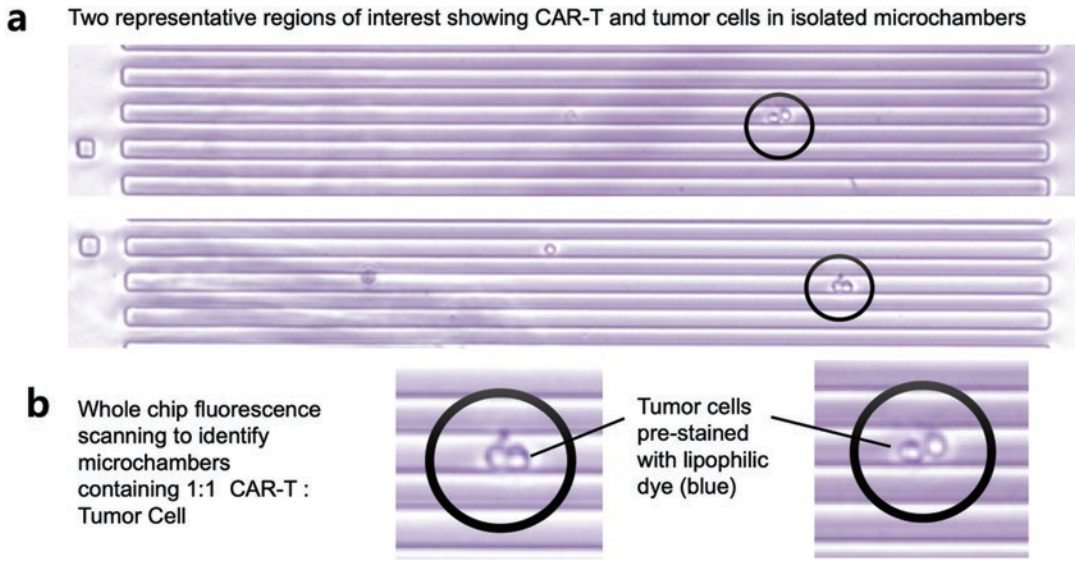

Fig. 2 Loading of CAR-T and tumor cells in microchambers for multiplex cytokine secretion assay. (**a**) Representative regions of the device showing the loading of CAR-T and/or tumor cells in isolated microchambers. (**b**) Enlarged view of 1:1 CAR-T:tumor cell pairs in microchambers. Tumor cells can be readily identified by prestaining with lipophilic dye (blue)

8. Top with the clamp plate and screw device together evenly (*see* **Note 40**).
9. Image chip with a microscope with automatic image stitching, and incubate at 37 °C for 14 h (*see* **Note 41**).
10. Submerge assembly into 5% BSA, disassemble the clamp plates, and carefully remove the barcode chip from the Cell Capture Microchamber Chip (*see* **Note 42**).
11. Rinse chip 5× with 1 mL of 5% BSA (*see* **Note 43**).
12. In a dark flat environment, add 400 μL of secondary antibody (1:200 dilution in 5% BSA), and incubate for 40 min (*see* **Note 31**).
13. Following incubation, repeat **step 10**.
14. In a dark flat environment, add 400 μL of streptavidin–fluorophore (1:100 dilution in 5% BSA), and incubate for 30 min (*see* **Note 31**).
15. Following incubation, repeat **step 10**.
16. In a dark flat environment, block chip with 5% BSA, and incubate for 30 min (*see* **Note 31**).
17. Wash with PBS, followed by (1:1) PBS/DI water, and then DI water (*see* **Note 32**).
18. Using filtered compressed air, blow the surface of the barcode chip dry (*see* **Note 34**).

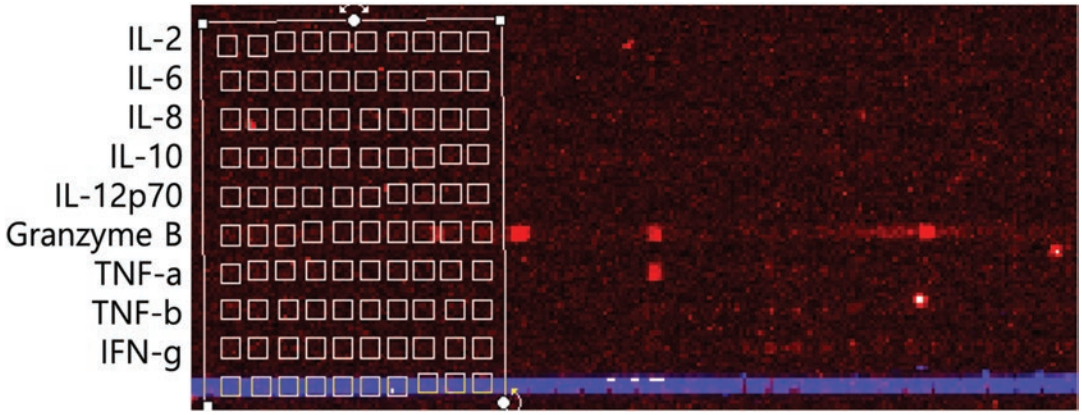

Fig. 3 Scanned fluorescence image for detection of single-cell or single-cell-pair cytokine sections. A square array is used in Genepix software to quantify protein signals. Each row represents a protein of interest and can be detected by fluorescent detection antibodies (red). A position reference (blue) is patterned to facilitate the identification of proteins. The microchambers (not shown) are perpendicular to the antibody lines

19. Scan chip using Genepix array scanner (Fig. 3) (*see* **Note 44**).
20. Combine Genepix array scanner with the positions of the microchambers and cell counts (Fig. 2).

4 Notes

1. This medium is dependent on the target cell culture media. For K562 and Raji cell lines, this is complete RPMI supplemented with 10% FBS.
2. For K562 and Raji cell lines, this is complete RPMI/RPMI supplemented with 10% FBS.
3. Obtained via Rong Fan Lab or CAD/Art Services, Inc., Engineering and Imaging Support Team.
4. PDMS Sylgard 184 (Dow Corning) can also be used.
5. Scotch brand magic tape is the only tape that does not leave a residue.
6. Blocking solution is best when prepared fresh.
7. Use the media that the cells were cultured with.
8. These should be different clones than those of the capture antibodies in order to prevent nonspecific signals.
9. We use the Nikon Eclipse Ti microscope.
10. The final culture medium of the CAR-T cells will be dictated by the target cell culture media.
11. For K562 and Raji cell lines, this step would be RPMI only (no FBS).

12. Quantity should be enough to cover approximately 3/4 of the wafer. Use the positive photoresist when making the flow-patterning master mold. Use the negative photoresist when making the cell capture microchamber mold.
13. If there are bubbles in the photoresist when pouring, the resulting pattern will have defects. Therefore, ensure there are no bubbles present.
14. Varying the spin speed and size of the wafer (i.e., 4″ in this protocol) changes the thickness of the photoresist. For technical specifications, check the photoresist manufacturer's spin speed curves (available here—http://www.microchem.com/PDFs_Dow/SPR%20220%20DATA%20SHEET%20R%26H.pdf).
15. Steady cool down is required to avoid cracking the photoresist.
16. To calculate the proper exposure time, use the mask aligner to measure lamp intensity, and divide this value by 150 mJ/cm^2.
17. It is best to fabricate an extra barcode flow-patterning chip in order to avoid delays in experiments.
18. Make sure the two agents are thoroughly mixed (at least 30 s) so that they cross-link and cure evenly.
19. Compressed air can be filtered by attaching a one-way filter to the outlet. Dust can be detrimental to the channels because it can cause the PDMS channels to connect, making them unusable.
20. Ensure boat is as close as possible to the wafer in order to discourage leakage and an uneven device.
21. To make sure that the chip has a uniform thickness, ensure that the oven surface is level. PDMS can also be left in overnight.
22. The silicon master mold is extremely delicate; therefore, be cautious. Wrapping the PDMS in plastic wrap while not in use helps keep it clean.
23. The holes should be created at the 20 inlet and 20 outlet templates in the pattern.
24. Without the complete removal of the liquid, the PDMS and glass slide will not be able to bond in subsequent steps.
25. Tweezers can also be used to apply force to the tape and PDMS.
26. If possible, perform this step in a clean room or hood to ensure this step is dust-free.
27. It is okay if there is debris in the channel; however, if the debris augments the shape, then it is not usable. Write down the

channels with deformities so that they're not used during flow patterning.

28. Make sure you mark the flow-patterned area for reference when attaching the single cell capture microchamber.
29. When loading antibodies, touch the pipette tip to the bottom of the inlet, and then, release the liquid. Otherwise, air bubbles can be introduced which can slow down the flow-patterning speed. Note the channel each antibody was placed in.
30. The PDMS bound to the glass slide can be delaminated if pressure is released too fast.
31. Petri dish covered in aluminum foil works well for this.
32. Remove these carefully as to avoid delaminating the chip.
33. Take care to avoid smearing antibody channels. Clean PDMS channels in a container of 1:1 IPA/DI water, and sonicate for 20–30 min. Discard IPA/DI water, refill with DI water, and sonicate again for 20–30 min. Prepare PDMS according to Subheading 3.3, beginning at **steps 10** and **11**.
34. Tweezers are helpful to avoid making contact with the barcode.
35. The chip can be stored for up to 2 weeks.
36. Target cells include Raji cells, K562, NIH 3T3, and ELM cell lines expressing CD19. Run this portion in triplicate, and incubate CAR-T and target cell controls in the same u-bottom 96-well plate.
37. Oxygen-plasma treatment decreases the PDMS hydrophobicity. Use the media that the cells are cultured with.
38. Ensure there is just a thin layer of media before adding cells. Following incubation, check that the cells have settled into the wells.
39. When laying the barcode chip down, align starting at the edge you placed the media along. Make sure that the chip is straight. However, once you have placed the barcode chip down, do not alter its position.
40. It is best to insert and tighten the screws in a diagonal fashion to ensure that an even amount of pressure is applied across the device.
41. We use the Nikon Eclipse Ti microscope.
42. Take care to avoid smearing antibody channels.
43. Dry the bottom of the chip with a Kimwipe, and place in a new petri dish.
44. Afterward, chip can be stored in a dark desiccator.

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